

Extracting hiking route descriptions from trajectory and terrain data

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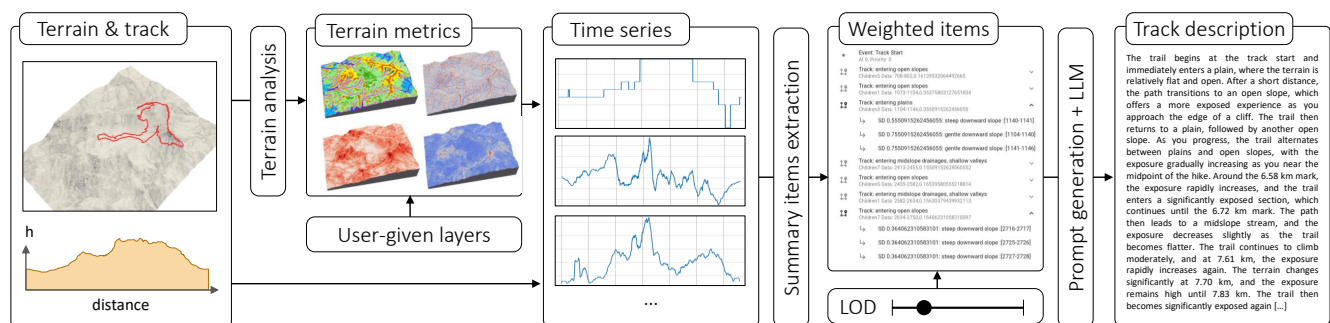


Figure 1: Given a terrain height field and a trajectory, our system computes a set of terrain-derived metrics that are then sampled along the trajectory. From these signals, we extract a collection of segments, descriptors, and events. These elements are then filtered according to their importance and a user-defined level of detail, before being finally translated into a natural-language description using an LLM.

Abstract

We consider the problem of analyzing a movement trajectory across a terrain in order to obtain a human-readable description of the route. Our input consists of a trajectory (e.g., a GPS track) and a digital elevation model of the area where the trajectory takes place. The proposed method starts by computing a set of terrain-derived metrics, such as a landform classification, slope, and exposure. Based on these signals, the trajectory is segmented into meaningful parts, such as those going through ascending slopes, flat areas, summits, or ridges, and their descriptive characteristics (e.g., steep, gentle, exposed). Finally, using these elements, we apply a Large Language Model (LLM) to transform the data into a natural language description of the route. The method is controlled by a single parameter that adjusts the number of relevant items in the trajectory, allowing users to easily obtain descriptions with varying levels of detail.

CCS Concepts

• **Computing methodologies** → Shape analysis; • **Human-centered computing** → Geographic visualization;

1. Introduction

Understanding and communicating movement through complex environments remains challenging in computer graphics and visualization. While modern acquisition techniques make it easy to capture dense spatial trajectories and high-resolution terrain data, conveying the essential characteristics of such data to users is still largely reliant on visual inspection, which can make it easy to overlook relevant elements unless they are explicitly highlighted. In addition, purely visual representations are not always suitable for downstream computational use: they are difficult to query, summa-

rize, or integrate into automated pipelines. This limitation becomes particularly apparent in the context of AI systems, which benefit from structured or linguistic representations of spatial information. Therefore, there is a need for methods that can distill geometric and layered information into concise, meaningful representations that are easier to interpret and share.

In this work, we explore an approach to semantic abstraction of spatial trajectories by combining terrain analysis with structured summarization and natural language generation. Given a track and its underlying terrain, we derive descriptive signals that capture rel-

event properties of the environment and the traversal. These signals are then aggregated into a hierarchical representation composed of segments, descriptors, and events, which together form a compact summary of the track. By design, this representation supports controllable levels of detail, allowing users to adjust the granularity of the output according to their needs, from brief overviews to more exhaustive descriptions. The signals and events that form the basis of our route descriptions can be considered *landmarks*, which aid the user in understanding the main features of the trajectory. In recent years, landmarks have been identified as a key component for wayfinding and navigation, both in urban and natural environments (see, e.g., [KIKS13, NT23]). In this context, a unique contribution of our method is that landmarks are extracted and selected automatically from the trajectory and terrain data. The generation of route descriptions in natural language has received some attention in the past, under the premise that there are many contexts where descriptions produced by humans, especially those that rely on salient features of the route, are preferable and easier to follow (see, e.g., [Bur00, RD08]). However, past research has focused mostly on the problem of generating route descriptions for navigation in road networks, where the network semantics, together with references like street names and landmarks, provides a natural framework to produce directions [DGP05]. To the best of our knowledge, generating hiking route descriptions has not been previously studied.

Our method enables the automatic production of textual descriptions of movement through 3D environments. This opens new possibilities for multimodal visualization, accessibility, and user interaction, where spatial data can be explored not only visually but also through language. Potential applications range from digital cartography and outdoor activity reporting to simulation analysis and virtual environment storytelling.

2. Method

A trajectory on a terrain is a spatial curve $\Gamma(t) : \mathbb{R} \rightarrow \mathbb{R}^3$ that represents a sequence of 3D positions over a parameter t . In practice, recorded trajectories, referred to as *tracks*, are typically available as discrete samples of such a curve. Although t does not necessarily correspond to physical time—for instance, in a hand-drawn trajectory—its unidirectional nature allows it to be interpreted as a temporal ordering. Following this analogy, we refer to additional properties and metrics defined along the track as *time series*, denoted by σ_i , where the value associated with $\Gamma(t)$ is given by $\sigma_i(t)$.

Figure 1 illustrates an overview of our system pipeline. The input consists of a track Γ and the corresponding terrain map $\mathcal{T}$, which is used both for visualization and for the automatic derivation of time series from terrain metrics (Section 2.1). We then aggregate the information provided by the different time series into a track summary, represented as a sequence of track segments, segment descriptors, and events (Section 2.2). The level of detail in this summary can be interactively controlled by the user via a slider, ranging from a highly concise description of the route to the full sequence of extracted elements. Finally, the filtered list is formatted into an appropriate prompt for a large language model (LLM), which generates the final natural-language description, optionally

following a user-specified narration style provided through a text field (Section 2.3).

2.1. Time series derivation

Input tracks are required to contain at least the sequence of coordinates along the trajectory. It is also common, particularly in hiking tracks, to annotate important landmarks or waypoints at specific positions. In such cases, we treat these waypoints as events (see next section). In recorded tracks, timestamps or relative time offsets since the start of the recording may also be available, although we cannot assume this information will be provided. Some recording devices might further provide additional monitored data, such as heart rate. In our application, however, we disregard temporal information and any user-specific data, focusing exclusively on generating narratives based on the trajectory's location data and the terrain characteristics. Nevertheless, such extra information could be incorporated to enable the generation of more subjective, first-person descriptions of a track.

Therefore, since the only required attribute of the input track Γ is its spatial coordinates, we derive additional time series using metrics that can be computed directly from the underlying terrain [AGSG25]. In our current implementation, for testing purposes, we consider the following time series:

- Path slope (σ_α). Given a unidirectional traversal of the track, we compute the path slope using forward differences of the elevation values.
- Terrain slope (σ_γ). Terrain slope is defined as the angle between the terrain gradient and the horizontal plane. This differs from path slope, as one may traverse inclined terrain without a net change in elevation, resulting in a zero path slope.
- Exposure (σ_{expo}). Intuitively, an exposed location corresponds to a position with a nearby steep drop. We approximated this by computing the difference between the elevation at a point and the minimum elevation within a local neighborhood of a fixed small radius.
- Landform class ($\sigma_{\mathcal{L}}$). Various landform classification schemes exist; we adopt the approach proposed by Weiss [Wei01], which categorizes terrain into ten classes (e.g., ridges, canyons, wide valleys, plains). This method relies on the Topographic Position Index (TPI), which measures the relative elevation of a point with respect to the mean elevation in a neighborhood. TPI is evaluated at two spatial scales, yielding nine possible combinations. Cases where TPI is close to zero at both scales are further subdivided into midslopes or flat areas based on a terrain slope threshold.

While we only included these layers for evaluation purposes of our framework, we note that there are many additional terrain properties potentially relevant for track description that can be automatically derived from the terrain elevation map. Examples include surface roughness, visibility index, or measures such as jut and rut, which aim to capture the impressiveness of a location. We refer the reader to the survey in [AGSG25] for a comprehensive overview of these and other metrics. Moreover, time series could also be obtained by sampling the track positions over existing user-provided terrain layers, such as land cover or vegetation density maps.

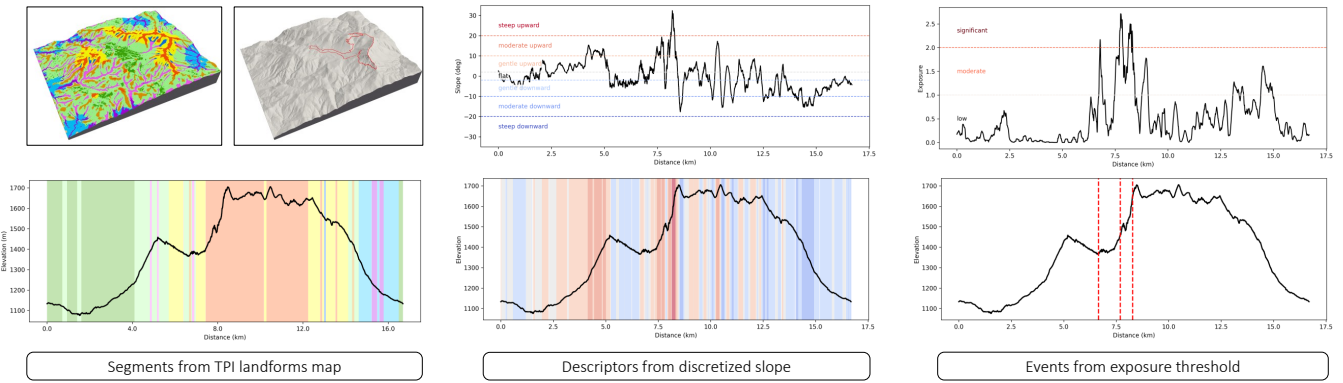


Figure 2: Example of track summary items derived from a trajectory. Segments are obtained from a discrete map (e.g., TPI landforms, left), descriptors are computed from discretized metrics (e.g., slope, center), and event generators (e.g., thresholded exposure, right) detect and annotate time points along the route.

2.2. Track summary

The derived time series are used to generate a set of *summary items*. We distinguish three types of items: track segments, segment descriptors, and events. Each item i is associated with an anchor point t_i along the track and a computed weight ω_i . These weights act as inverse priorities for inclusion in the final filtered summary: by sorting all items according to their weight, from smallest to largest, we can retain any desired proportion of the most relevant elements.

Track segments s_i are obtained by partitioning the track according to the discrete values of a selected time series. As track segments represent the primary structural unit of the final summary, the chosen series should be the one that captures the most relevant aspect of the trajectory. In our implementation, we chose σ_L so to base our narration on the type of landform being traversed (Figure 2, left). Land cover maps could also lead to interesting top-level segmentations, or forest type if the users wants to shift the narration focus on ecological characteristics, for example. Formally, we create a segment for every maximal interval $[t_a, t_b]$ over which $\sigma_L(t)$ remains constant. The associated anchor point for each segment is its starting position t_a . The priority of a track segment is determined by both the significance of the transition from the preceding value to the current one in the time series (e.g., transitioning from plains to ridges is more significant than transitioning from midslopes to upper slopes), as well as the length of the segment.

Segment descriptors $d_{i,j}$ are obtained by further partitioning each track segment s_i according to an additional discrete time series. Multiple descriptors can be applied independently and in parallel. Conceptually, if a track segment represents a noun (i.e., the type of terrain being traversed), segment descriptors act as adjectives that refine its description and provide additional details. In our implementation, we used σ_α , σ_γ and σ_{expo} . Since these metrics are continuous, we discretize them into categories. For example, we can categorize slopes as flat, gentle, moderate, or steep (Figure 2, center); and exposure as none, moderate, or significant. Each descriptor is anchored at its starting point, which lies within the interval $[t_a, t_b]$ of the corresponding segment. To ensure that a segment

descriptor $d_{i,j}$ is only selected if its associated segment s_i is also included, we enforce that its priority is always lower. We achieve this by adding the weight of s_i to the weight computed for $d_{i,j}$.

Events e_i represent significant moments along the track and are anchored to individual time points t_i . They are typically detected by thresholding a time series (Figure 2, right). Note that the same time series can be used both to generate segment descriptors and to detect events. While this might appear redundant, their roles differ: events are independent of track segments and capture isolated points in the track chronology. For example, a descriptor such as "significantly exposed" may characterize a segment, so if the segment is not relevant enough to appear in the filtered summary, the descriptor will not be listed either. In contrast, exceeding a more extreme exposure threshold may instead trigger a standalone event, which is reported as a separate sentence in the final summary. Because of this, events typically are assigned low weights (high priority). As mentioned earlier, route waypoints can also be incorporated as events, allowing the system to signal the traversal of user-defined points of interest along the track.

2.3. Narrative generation

We leverage the capabilities of modern LLMs to transform the extracted summary into a cohesive, human-readable narrative.

To control the length and density of the generated text, we use a budget-based filtering scheme. Through an interactive slider, the user specifies the percentage of summary items to retain from the full set. We then sort items by their weight w_i and keep the n lowest-weight elements. The selected subset is subsequently reordered by their anchor position t_i to ensure a coherent chronological progression. The retained items are converted into a structured intermediate representation in the form of bullet points. We use distinct templates for each category: track segments and descriptors are expressed as spatial intervals (e.g., "from {start} to {end} along the track, there is a {name}"), while events are anchored to specific positions. This representation is exposed to the user for inspection before being passed to the LLM.

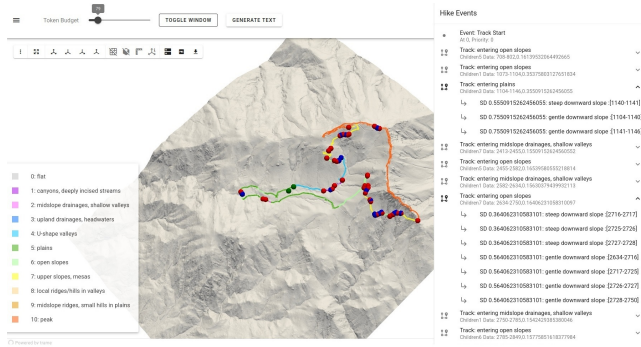


Figure 3: Screenshot of the user interface. The filtering of summary elements is interactively reflected in the 3D view and the items list.

Finally, the validated bullet list is injected into a predefined prompt containing contextual and stylistic instructions (see supplementary material for prompt details). By adjusting the prompt template or selecting alternative models, users can tailor the output to a range of styles, from concise technical summaries to more expressive, narrative descriptions.

3. Results

We implemented our application using Python and Trame for a web-based viewer and interaction (Figure 3). We intend to make our implementation publicly available in the near future.

Upon loading a digital elevation model (DEM), the system computes the corresponding terrain-derived metric maps. While our current implementation relies on a fixed set of metrics for evaluation, we plan to extend it to support user-provided layers, as well as to allow users to select which automatically derived metrics should be used as descriptors. Once a track is loaded, the system samples the associated time series and generates the full set of prioritized summary items. A slider enables users to adjust the level of detail of the resulting bullet-point list. An interactive 3D viewer displays the terrain and the trajectory, allowing users to visually inspect the segments and events selected in the summary.

The final narrative is generated via the Hugging Face API, enabling flexible experimentation with different large language models. In particular, we have conducted our experiments using Qwen/Qwen2.5-7B-Instruct. To avoid unnecessary calls to the language model, narrative generation is triggered on demand, using the current selection of items. The user can further control the narrative style through a text input field.

The following text presents an example generated narrative, corresponding to the track shown in Figure 3:

The trail begins at the track start and quickly leads you into a midslope stream, where the terrain remains relatively gentle. After a short distance, you ascend into an open slope, where the exposure to the edge begins to increase. The trail then follows another midslope stream, before you encounter a second open slope with increasing exposure. Further along,

the trail enters a ridge, where the exposure rapidly increases, and the terrain becomes more challenging. The ridge section is followed by another open slope, and then a midslope stream. The exposure continues to rise as you approach the next section of open slope. At 6.65 kilometers, the exposure increases sharply, and the trail becomes more exposed. The terrain then changes dramatically as you encounter several changes in slope, each adding to the exposure. The trail continues through an upper slope and then reaches a peak, where the exposure remains high. You then traverse another upper slope and headwaters before reaching another peak. The trail then leads you to a ridge, followed by an open slope. As you progress, the trail descends into a wide valley, then a deep stream, and finally into a plain. The exposure remains high throughout the latter part of the trail, especially as you approach the final section. The track ends at 16.50 kilometers, where the terrain levels off into a plain.

The evaluation procedure followed was qualitative, based on visual inspection of the input, the summary items, and the generated narrative. Our goal was simply to ensure that the description read natural and contained the (filtered) summary items, but we did not attempt to evaluate the accuracy or faithfulness of the description—something clearly relevant, but that is left for future work.

As can be observed, the generated narrative is cohesive and employs appropriate natural-language transitions between segments (e.g., “The terrain then changes”, “The trail continues”). It is also able to merge multiple bullet points into more fluid sentences (e.g., “the trail descends into a wide valley, then a deep stream, and finally into a plain.”) and in some cases it even summarizes several descriptors (e.g., “you encounter several changes in slope”). Overall, this demonstrates that our pipeline is effective in producing coherent descriptions from only a terrain elevation map and a trajectory. However, some sentences feel unnatural or repetitive (see, for example, the passage from “The exposure continues to rise” to “adding to the exposure”). This issue likely arises from the selection of bullet points, where excessive importance was probably given to exposure descriptors in our manually-tuned weight assignment criteria. Moreover, the landform classification we used originally associated multiple semantic labels for some of the classes (e.g., a high small-scale TPI and near-zero large-scale TPI combination represents either midslope ridges or small hills in plains). We prescribed only one of the names per class, but without additional context this can lead to ambiguous or confusing descriptions. Finally, the generation of summary items relies on thresholding the time series. In the presence of noisy signals, this may produce an excessive number of items. Although the priority-based filtering alleviates this issue to some extent, exploring more robust filtering strategies would likely improve the overall quality of the generated summaries.

For now, we mitigate these issues directly at the LLM level by explicitly prompting it to connect the items as “*natural description in paragraph format*” and mentioning “*you may collapse multiple points to single sentences.*”. Due to space limitations, we refer the reader to the supplementary material for several examples of generated narratives that used our updated prompt.

Finally, regarding narrative style, we observed only minor variations in the generated text using the LLM model from our pipeline. However, when providing the same formatted prompt to ChatGPT, using their website, we obtained results that were more aligned with

our expectations. This suggests that the observed limitations are likely due to the specific model employed rather than to shortcomings in our pipeline.

4. Conclusion

We have presented our preliminary results on an automated system for generating textual descriptions of a track. Our experiments indicate that terrain-derived metrics and landform classifications sampled along the track are effective in capturing relevant aspects of the route, which then an LLM can use to create the narrative. As future work, we plan to address the limitations discussed above. In particular, given a corpus of route narratives, we could aim to learn data-driven weights for the different metrics and landform transitions, enabling more natural priorities. We also believe that richer descriptions could be obtained by detecting whether the trajectory crosses or follows terrain features (e.g., streams or ridges), as this distinction can significantly influence the interpretation of the route. Regarding this last point, we would like to conduct a user experiment to assess if our generated narratives are understandable and meaningful to human readers.

Acknowledgments

This work was supported by grant PID2021-122136OB-C21 from MCIN/AEI/10.13039/501100011033/FEDER, grant PID2023-150725NB-I00 from MICIU/AEI/10.13039/501100011033, and by grant “ALECTORS 2024” from Universitat Politècnica de Catalunya.

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